Method Overview: Project Subtype and Continuous Score in SSC

1. Task Definition

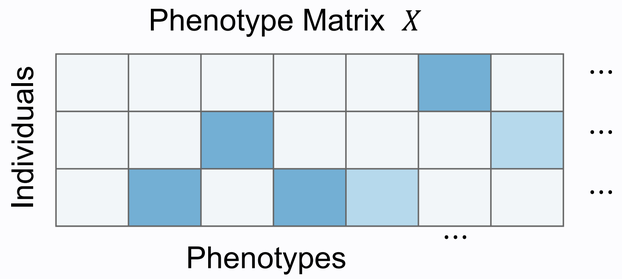
Input:

- SPARK → Phenotype, Subtype, Subtype score
- SSC → Phenotype

Output:

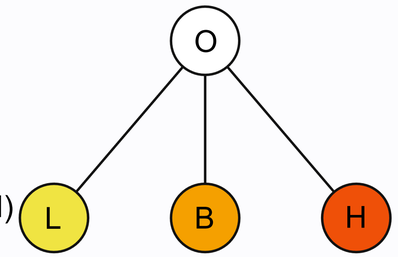
- SSC → Subtype (L/B/H), Subtype score

2. Reference SPARK



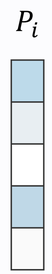
3. Model: DDRTree

- Root node: O
- Three subtype nodes:
  - ✓ L: Low-support-needs (**LSN**)
  - ✓ B: Emotional-behavioral-support-needs (**EBSN**)
  - ✓ H: High-support-needs (**HSN**)



4. SSC individual processing (for each SSC individual i)

4.1 Raw phenotypes



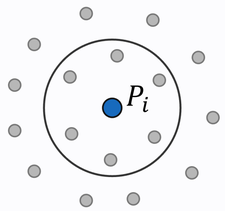
4.2 Reference-based scaling

(e.g., mean/std)



4.3 KNN in scaled space

$N_i = KNN(P_i, X_s)$



Find K nearest neighbors in SPARK

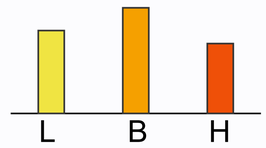
4.4 Neighbor information

$\{S_j, d_j, \ldots\}$

S_1 (L)
S_2 (B)
S_3 (H)

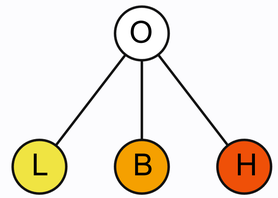
Collect neighbors' subtype and distance info

4.5 Count neighbors per node $n_i(u)$



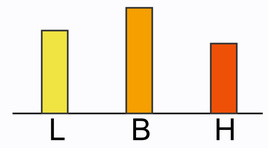
Number of neighbors mapped to each subtype node u

4.6 Subtype-specific tree distance $D_i(u)$



Compute weighted mean root-distance to each node u

4.7 Normalize by subtype scale $R_i(u)$



Normalize $D_i(u)$ by subtype scale

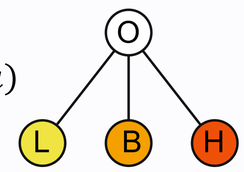
5. Convert to Similarity $P_i(u)$



Covert $R_i(u)$ to similarity to each subtype node u

6a. Subtype Projection

$\hat{z}_i = \text{argmax}_{u \in \{L,B,H\}} P_i(u)$



Projected subtype as the node with highest similarity

Projected subtype: *B*

6b. Subtype score Projection

$$\hat{c}_i = \sum_{u \in \{L,B,H\}} P_i(u) c_u$$

Projected subtype score as similarity-weighted mean of subtype scores

Node(u)	L	B	H
c_u	c_L	c_B	c_H

$\hat{c}_i$ = weighted mean of c_u

Detailed explanation of $n_i(u)$

For each subtype node $u \in \{L, B, H\}$: $n_i(u)$ is the number of SPARK neighbors among the K nearest neighbors of SSC individual i that are mapped to node u in the DDRTree.

- $n_i(u) \geq 0$
- $\sum_{u \in \{L,B,H\}} n_i(u) = K$

Notation

P_i	: Phenotype vector of SSC individual i
X	: Phenotype matrix of SPARK
X_s	: Selected phenotype matrix of SPARK
N_i	: Set of K nearest neighbors in SPARK
S_j	: Subtype of neighbor j (L/B/H)
d_j	: Distance from P_i to neighbor j
$n_i(u)$	: Number of neighbors mapped to u
$D_i(u)$	: Weighted mean root-distance to u
$R_i(u)$	: Normalized distance for node u
$P_i(u)$	: Similarity of i to node u
$\hat{z}_i$	: Projected subtype (L/B/H)
$\hat{c}_i$	: Projected subtype score

Key Idea: Leverage a reference DDRTree learned from SPARK to project SSC individuals to subtypes based on neighbor information in phenotype space, and project subtype subtype score via similarity-weighted averaging.

LSN EBSN HSN